Test Case 6 – Loan Eligibility & Installment Calculation

Request Data

Declared Monthly Installments: 6,000

Declared Net Profit: 520,000

Declared Business Revenue: 5,000,000

Requested Loan Amount: 250,000

Requested Tenor: 36 months

Date of Birth: 11/01/1983

Bank Statement Credit (Total Income): 4,750,000

SIMAH Installment Loan Balances: 325,000

SIMAH Revolving Credit Limits: 30,000

SIMAH Monthly Installments: 6,000

Step 1: Maximum Tenor Calculation

monthsTo70 = 330 months

riskGradeMaxTenor = 36 months

maxTenor = MIN(330, 36) = 36 months

Step 2: Effective Revenue

Effective Revenue = MIN(Declared Business Revenue, Bank Statement Credit)

= MIN(5,000,000, 4,750,000) = 4,750,000

Step 3: Exposure Calculation

Total Exposure = SIMAH Installment Loan Balances + SIMAH Revolving Credit Limits

= 325,000 + 30,000 = 355,000

Exposure Percentage = (355,000 / 4,750,000) × 100 = 7.47%

Step 4: Eligible Loan Amount Calculation

MaxByRevenue = Effective Revenue × 15% = 4,750,000 × 0.15 = 712,500

Available Exposure Limit = (Effective Revenue × 30%) " Total Exposure

= (4,750,000 × 0.30) " 355,000 = 1,070,000

Second Cut Limit = MIN(712,500, 1,070,000) = 712,500

Max Loan Limit = 500,000

Eligible Loan Amount = MIN(500,000, 712,500) = 500,000

Step 5: Monthly Net Profit

Bank Statement Credit × 15% = 4,750,000 × 0.15 = 712,500
MIN(712,500, 520,000) = 520,000
Monthly Net Profit = 520,000 ÷ 12 = 43,333.33

Step 6: Disposable & Available Disposable Income
Disposable Income = monthlyNetProfit * 0.5
= 43,333.33 × 50% = 21,666.67
Monthly Obligations = SIMAH Monthly Installments + (5% × SIMAH Revolving Credit Limits)
= 6,000 + (30,000 × 0.05) = 7,500
Available Disposable Income = disposableIncome - monthlyObligations
= 21,666.67 " 7,500 = 14,166.67

Step 7: Installment Calculation
Annual Reducing Rate = 14.57%
Monthly Rate = 14.57 ÷ 100 ÷ 12 = 0.0121416667
First-Cut Installment = P × r / 1" (1 + r)" n

Where
n = eligibleTenor
P = eligibleLoanAmount

After apply all there values to formula
First-Cut Installmen = 17,227.56

Step 8: Maximum Affordable Installment
MIN(monthlyNetProfit * 0.25, availableDisposableIncome)
25% of Monthly Net Profit = 43,333.33 × 0.25 = 10,833.33
MIN(10,833.33, 14,166.67) = 10,833.33
Maximum Affordable Installment = 10,833.33

Step 9: Final Eligibility Decision

if(firstCutInstallment<maxInstallment)
finalEligibleLoanAmount = eligibleLoanAmount;
finalInstallmentAmount = firstCutInstallment; (
else

finalInstallmentAmount = maxInstallment;
finalEligibleLoanAmount = maxInstallment × (1 - (1 + monthlyRate)^(-tenor)) / monthlyRate;

the calculation proceeds to the affordability adjustment.
Final Installment Amount = 10,833.33
Final Eligible Loan Amount = 314,418.53

Step 10: Calculate Final Loan Amount and EMI based on new eligible amount and Sent to LMS
Final Loan Amount = MIN(Final Eligible Loan Amount, Requested Loan Amount)
= MIN(314,418.53, 250,000) = 250,000
Final EMI = (P × monthlyRate) / (1 " (1 + monthlyRate)^(-n))
monthlyRate = annualReducingRate / 12
n = eligibleTenor
P = finalLoanAmount
After applying these values
Final EMI = 8613.78

Final Outcome:
Risk Grade: A,
Final Eligible Loan Amount: 250000,
Final Installment Amount: 8613.78,
First-Cut EMI: 17227.56,
Max Affordable EMI: 10833.33,
Annual Reducing Rate: 14.57,
Eligible Tenor: 36,
Maximum Tenor: 36,
Exposure %: 7.47,
Monthly Net Profit: 43333.33,
Monthly Disposable Income: 21666.67,
Available Disposable Income: 14166.67